



UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO
FACULTAD DE INGENIERÍA
DIVISIÓN DE INGENIERÍA ELÉCTRICA
INGENIERÍA EN COMPUTACIÓN
COMPUTACIÓN GRÁFICA e INTERACCIÓN HUMANO
COMPUTADORA



DOCUMENT: User manual

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Theory Group: 05

Semester: 2026-1

Deadline: November 24, 2025

Objective statement

- Develop an interactive virtual tour in OpenGL 3
Create a virtual tour that recreates a front side and two rooms (with their elements and artistic style), integrates a synthetic camera and four animations, and includes all technical and user documentation (in Spanish and English), a cost analysis, and a functional executable.
- Select and document references
Collect and include in the delivery PDF at least three reference images for each room and the front side (real or fictional, excluding the specified locations).
- Recreate 5 elements per room with consistent style
Model and integrate five objects per room, ensuring that the artistic style of each element matches the references.
- Implement a synthetic camera
Program a free/controlled camera that allows navigation through the tour (movement, rotation, and zoom).
- Develop 4 animations
Implement four distinct animations (e.g., a door opening, an object oscillating, a spotlight effect, a rotating object animation).
- Build complete and bilingual documentation
Generate: Gantt chart, user manual (objectives, technologies, interaction), technical manual (architecture, shaders, transformations), all in Spanish and English (translations reviewed).
- Perform a cost analysis
Document project costs (hours, software, licenses, resources) and propose a justified sale price.
- Generate executable and GitHub repository
Compile and package a functional executable (not the debug folder), upload the project to GitHub using relative paths, without compressed files or files larger than 100 MB.
- Ensure realism and visual quality
Adjust lighting, materials, and layout to maximize the realism of the space

(evaluated according to course criteria).

- Ensure the project is individual
Confirm that the work is completely individual or, if using previous team work, create and furnish a new room with original elements.
- Submit on time and in digital format
Upload the PDF and executable to Classroom before the deadline and keep the GitHub repository accessible.

1 Tour controls

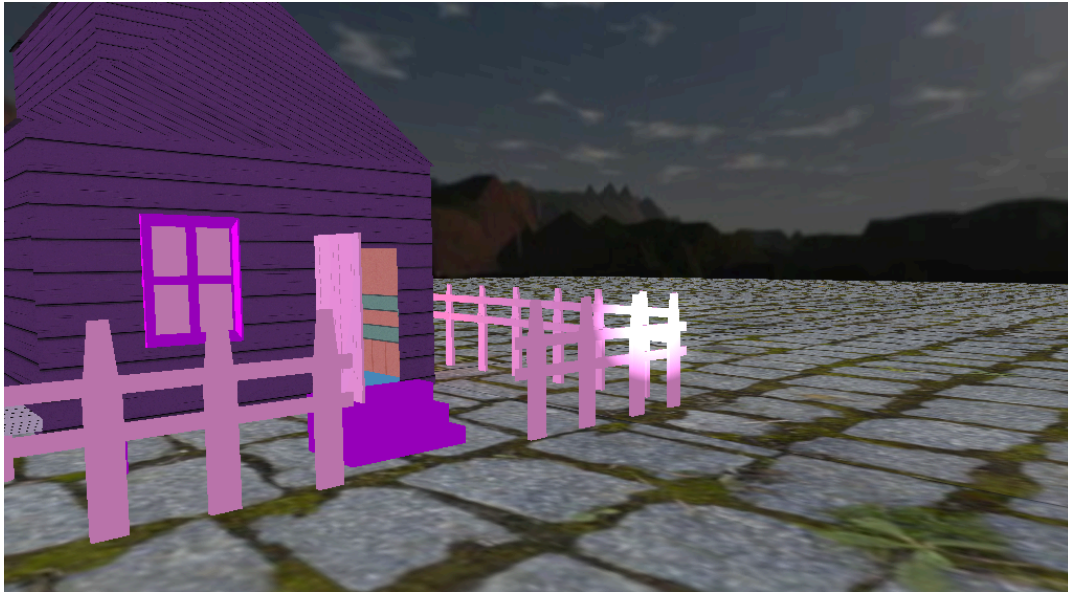
- **Synthetic camera**

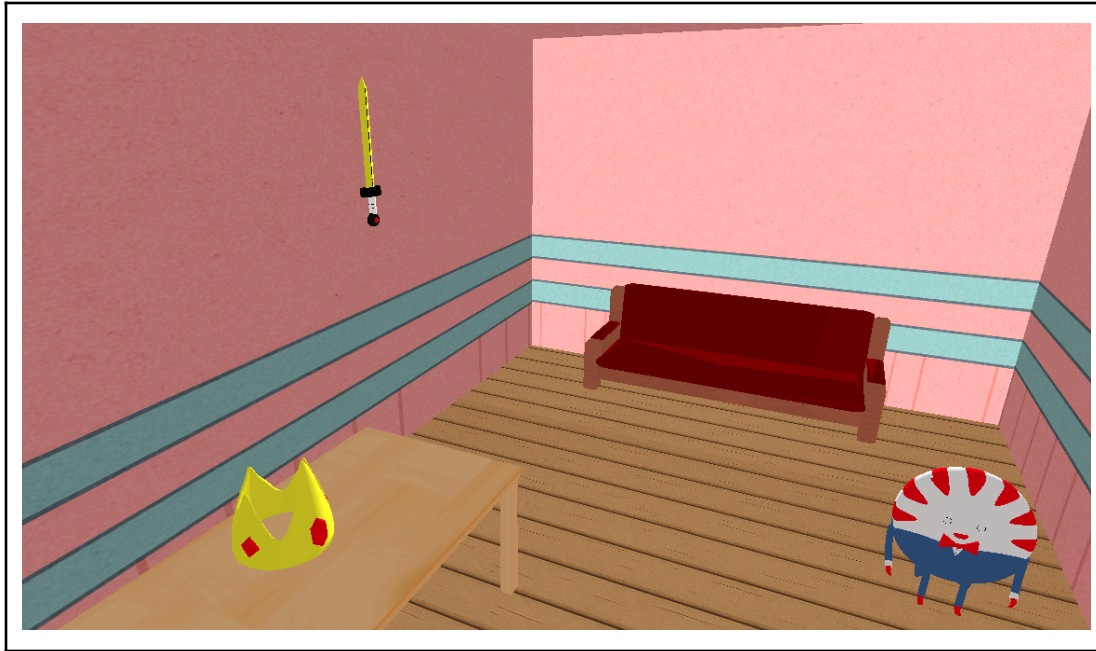
Allows freely exploring the elements from any angle and position.

How is it activated?
When running the program, the default active camera is the free camera.
How is it used?
Mouse: Allows changing the direction the camera is pointing. Keyboard: Allows moving the camera in different directions. • [W] Move forward • [S] Move backward

- [A] Move left
- [D] Move right

Example of use

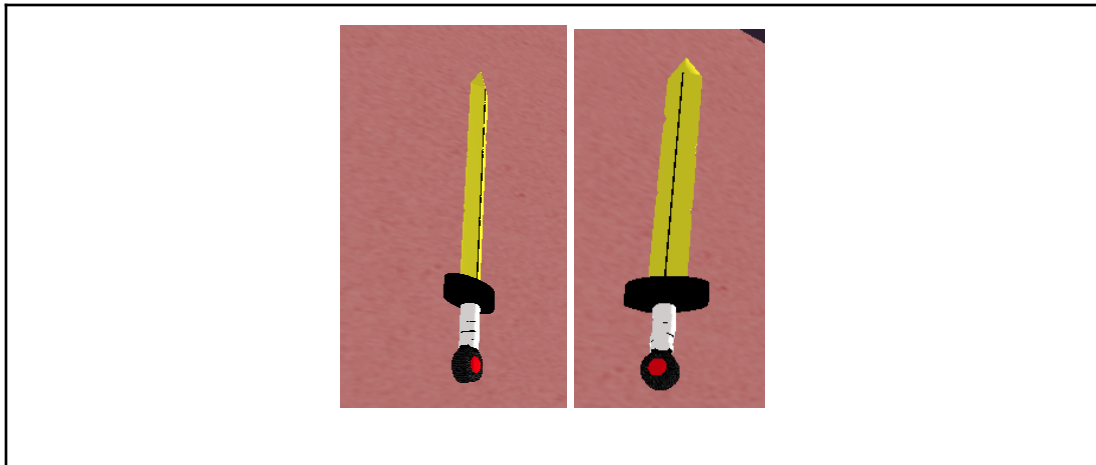




2 Animation controls

- **Sword levitation**

How is the animation executed?
The default animation runs in a loop.
Example of use



- **Door opening**

How is the animation executed?
Press [M] to open the door at any time. Press [N] to close the door at any time.
Example of use
Two side-by-side images of a purple door. The left image shows the door closed with a glowing yellow circle in the center. The right image shows the door open, revealing a room with a blue floor and a small character standing in the doorway.

3 List of software, tools, and IT to be used

- **Graphics Drawing API**
OpenGL 3.3+ will be used as the graphics drawing API, since it is a project requirement and I am also familiar with its use.
- **Interface Library**
GLFW is planned to be used due to familiarity with it and its use in the laboratory.
- **Texturing Library: stb_image, SOIL, etc.**
For texturing, stb_image is planned to be used because it is easy to use and has extensive documentation available.
- **Model Loading Library**
ASSIMP will be used for model loading, as it is the library currently used in the laboratory.
- **Modeling Program**
Blender was chosen as the modeling program because it is the software I am most familiar with.
- **Image Editing Program**
For image editing, GIMP was chosen since it is free and highly compatible with different file formats.
- **Version Control System**
A public GitHub repository will be used to manage version control for the project.
- **Audio Library**
Although this aspect is still pending, after some research, OpenAL or FMOD is proposed for audio implementation.

4 Cost analysis

Direct costs

Concept	Quantity	Unit cost	Total
Development hours (OpenGL programming, modeling,	45 h	\$120 MXN/h	\$5,400 MXN

animations)			
Documentation preparation (technical manual, user manual, English/Spanish, Gantt chart)	10 h	\$120 MXN/h	\$1,200 MXN
Rendering / model adjustments	5 h	\$120 MXN/h	\$600 MXN
Electricity used during development (estimated)	25 horas de uso del equipo	\$4 MXN/h	\$100 MXN
External resources (textures, free models with no monetary cost but requiring adaptation time)	3 h	\$120 MXN/h	\$360 MXN

Indirect costs

Concept	Description	Total
Depreciation of computer equipment	Proportional use of equipment (PC, GPU) during the project	\$300 MXN
Software used	IDE, Blender, repositories, free tools with operational costs	\$0 MXN (free software)
Internet	Proportion of the monthly payment (10% of the \$400 MXN monthly plan)	\$40 MXN

Direct cost (\$7,660 MXN) + Indirect cost (\$340 MXN) = \$8,000 MXN

4 Bibliography

- Holubov, P. (2020, October 1). *Vintage Retro Radio* [3D model]. Sketchfab. <https://sketchfab.com/3d-models/vintage-retro-radio-26fb0de1015d4570bac28cd8b1fb0a62#download>
- Unknown author. (n.d.). *Marceline's AX Bass* [3D model]. Sketchfab. <https://sketchfab.com/3d-models/marcelines-ax-bass-417178d709774642b0d5b5a02181caa2>
- Mendes, O. (n.d.). *Ice King* [3D model]. Sketchfab. <https://sketchfab.com/models/998ea9be02194e618ae3551d36764f1d>
- luisguilhermefoliveira. (2021, August 7). *Sword Finn* [3D model]. Sketchfab. <https://sketchfab.com/3d-models/sword-finn-8ffcf2736553498ead2d8634ac6d574c>
- Unknown author. (n.d.). *Mesa da hora de madeira* [3D model]. Sketchfab. <https://sketchfab.com/3d-models/mesa-da-hora-de-madeira-446665edffa24c5f806a8ab507682b85>
- Unknown author. (n.d.). *BMO* [3D model]. Sketchfab. <https://sketchfab.com/3d-models/bmo-0821dc59eedd469db946b8adb202af11>